

Homework 1

Please complete the following problems. You must submit the following for each problem: a) the answer you are asked to return and b) Python code that can be run to verify your work.

1. Counting DNA Nucleotides

Making up all living material, the cell is considered to be the building block of life. The nucleus, a component of most eukaryotic cells, was identified as the hub of cellular activity 150 years ago. Viewed under a light microscope, the nucleus appears only as a darker region of the cell, but as we increase magnification, we find that the nucleus is densely filled with a stew of macromolecules called chromatin. During mitosis (eukaryotic cell division), most of the chromatin condenses into long, thin strings called chromosomes. See Figure 1 for a figure of cells in different stages of mitosis.

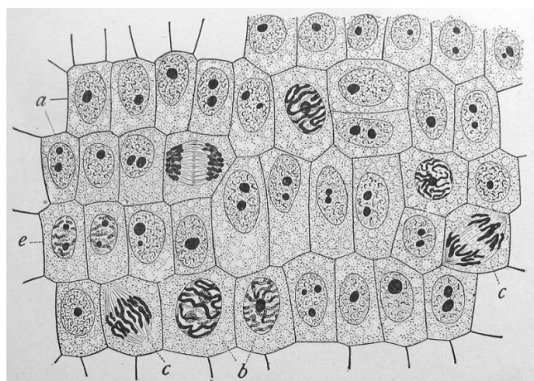


Figure 1: A 1900 drawing by Edmund Wilson of onion cells at different stages of mitosis. The sample has been dyed, causing chromatin in the cells (which soaks up the dye) to appear in greater contrast to the rest of the cell.

One class of the macromolecules contained in chromatin are called nucleic acids. Early 20th century research into the chemical identity of nucleic acids culminated with the conclusion that nucleic acids are polymers, or repeating chains of smaller, similarly structured molecules known as monomers. Because of their tendency to be long and thin, nucleic acid polymers are commonly called strands.

The nucleic acid monomer is called a nucleotide and is used as a unit of strand length (abbreviated to nt). Each nucleotide is formed of three parts: a sugar molecule, a negatively charged ion called a phosphate, and a compound called a nucleobase ("base" for short). Polymerization is achieved as the sugar of one nucleotide bonds to the phosphate of the next nucleotide in the chain, which forms a sugar-phosphate backbone for the nucleic acid strand. A key point is that the nucleotides of a specific type of nucleic acid always contain the same sugar and phosphate molecules, and they differ only in their choice of base. Thus, one strand of a nucleic acid can be differentiated from another based solely on the order of its bases; this ordering of bases defines a nucleic acid's primary structure.

For example, Figure 2 shows a strand of deoxyribose nucleic acid (DNA), in which the sugar is called deoxyribose. The only four choices for nucleobases are molecules called adenine (A), cytosine (C), guanine (G), and thymine (T).

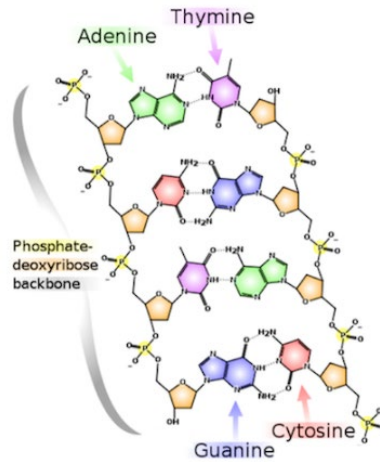


Figure 2: A sketch of DNA's primary structure.

DNA is found in all living organisms on Earth, including bacteria; it is even found in many viruses (which are often considered to be nonliving). Because of its importance, we reserve the term genome to refer to the sum total of the DNA contained in an organism's chromosomes.

PROBLEM

A string is simply an ordered collection of symbols selected from some alphabet and formed into a word; the length of a string is the number of symbols that it contains.

An example of a length 21 DNA string (whose alphabet contains the symbols 'A', 'C', 'G', and 'T') is "ATGCTTCAGAAAGGTCTTACG."

Given: A file containing a DNA string s of length at most 1000 nt.

Return: Four integers (separated by spaces) counting the respective number of times that the symbols 'A', 'C', 'G', and 'T' occur in s .

Sample Dataset

AGCTTTTCATTCTGACTGCAACGGGCAATATGTCTCTGTGTGGATTAAAAAAGAGTGTCTGATAGCAGC

Sample Output

20 12 17 21

2. Transcribing DNA into RNA

A second nucleic acid exists alongside DNA in the chromatin; this molecule, which possesses a different sugar called ribose, came to be known as ribose nucleic acid, or RNA. RNA differs further from DNA in that it contains a base called uracil in place of thymine; structural differences between DNA and RNA are shown in Figure 1. Biologists initially believed that RNA was only contained in plant cells, whereas DNA was restricted to animal cells. However, this hypothesis dissipated as improved chemical methods discovered both nucleic acids in the cells of all life forms on Earth.

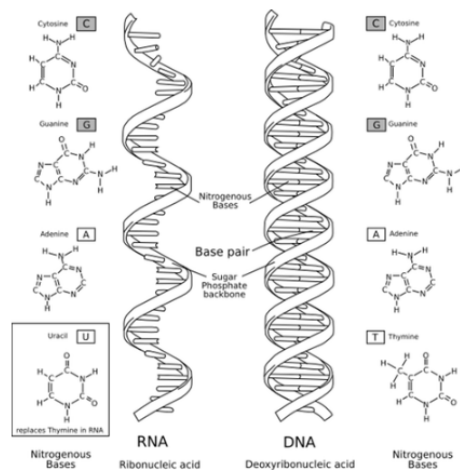


Figure 1: Structural differences between RNA and DNA.

The primary structure of DNA and RNA is so similar because the former serves as a blueprint for the creation of a special kind of RNA molecule called messenger RNA, or mRNA. mRNA is created during RNA transcription, during which a strand of DNA is used as a template for constructing a strand of RNA by copying nucleotides one at a time, where uracil is used in place of thymine.

In eukaryotes, DNA remains in the nucleus, while RNA can enter the far reaches of the cell to carry out DNA's instructions. In future problems, you will examine the process and ramifications of RNA transcription in more detail. One of the most important features of any programming language is its ability to manipulate variables. A variable is just a name that refers to a value; you can think of a variable as a box that stores a piece of data.

PROBLEM

An RNA string is a string formed from the alphabet containing 'A', 'C', 'G', and 'U'.

A DNA string t corresponding to a coding strand, its transcribed RNA string u is formed by replacing all occurrences of 'T' in t with 'U' in u .

Given: A file containing a DNA string t having length at most 1000 nt.

Return: The transcribed RNA string of t .



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Bioinformatics Engineering

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Sample Dataset

GATGGAACTTGACTACGTAAATT

Sample Output

GAUGGAACUUGACUACGUAAAUU

3. Complementing a Strand of DNA

The primary structure of a nucleic acid is determined by the ordering of its nucleobases along the sugar-phosphate backbone that constitutes the bonds of the nucleic acid polymer. Yet primary structure tells us nothing about the larger, 3-dimensional shape of the molecule, which is vital for a complete understanding of nucleic acids.

The search for a complete chemical structure of nucleic acids was central to molecular biology research in the mid-20th Century, culminating in 1953 with a publication in *Nature* of fewer than 800 words by James Watson and Francis Crick. Consolidating a high-resolution X-ray image created by Rosalind Franklin and Raymond Gosling with several established chemical results, Watson and Crick proposed the following structure for DNA:

1. The DNA molecule is made up of two strands, running in opposite directions.
2. Each base bonds to a base in the opposite strand. Adenine always bonds with thymine, and cytosine always bonds with guanine; the complement of a base is the base to which it always bonds; see Figure 1.

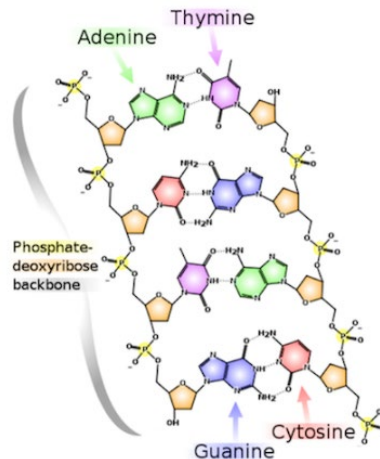


Figure 1: Base pairing across the two strands of DNA.

3. The two strands are twisted together into a long spiral staircase structure called a double helix; see Figure 2.

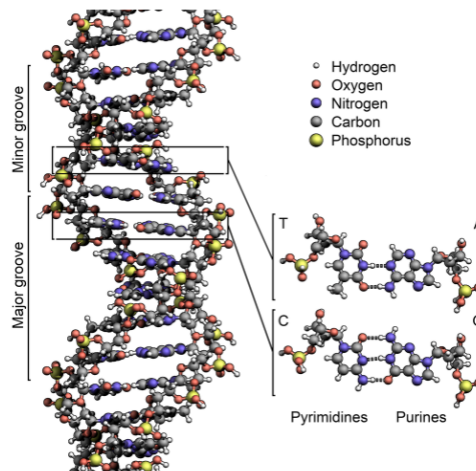


Figure 2: The double helix of DNA on the molecular scale.



Because they dictate how bases from different strands interact with each other, (1) and (2) above compose the secondary structure of DNA. (3) describes the 3-dimensional shape of the DNA molecule, or its tertiary structure.

In light of Watson and Crick's model, the bonding of two complementary bases is called a base pair (bp). Therefore, the length of a DNA molecule will commonly be given in bp instead of nt. By complementarity, once we know the order of bases on one strand, we can immediately deduce the sequence of bases in the complementary strand. These bases will run in the opposite order to match the fact that the two strands of DNA run in opposite directions.

PROBLEM

In DNA strings, symbols 'A' and 'T' are complements of each other, as are 'C' and 'G'.

The reverse complement of a DNA string s is the string s^c formed by reversing the symbols of s , then taking the complement of each symbol (e.g., the reverse complement of "GTCA" is "TGAC").

Given: A file containing the DNA string s of length at most 1000 bp.

Return: The reverse complement s^c of s .

Sample Dataset

AAAACCCGGT

Sample Output

ACCGGGTTT

4. Translating RNA into Protein

Just as nucleic acids are polymers of nucleotides, proteins are chains of smaller molecules called amino acids; 20 amino acids commonly appear in every species. Just as the primary structure of a nucleic acid is given by the order of its nucleotides, the primary structure of a protein is the order of its amino acids. Some proteins are composed of several sub-chains called polypeptides, while others are formed of a single polypeptide; see Figure 1.

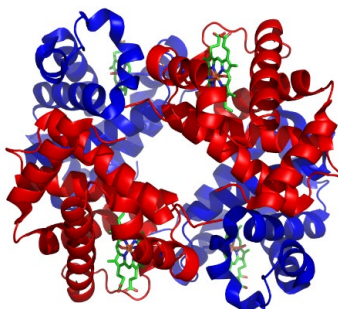


Figure 1: The human hemoglobin molecule consists of 4 polypeptide chains; α subunits are shown in red, and β subunits are shown in blue.

Proteins power every practical function carried out by the cell, and so presumably, the key to understanding life lies in interpreting the relationship between a chain of amino acids and the function of the protein that this chain of amino acids eventually constructs. Proteomics is the field devoted to the study of proteins.

How are proteins created? The genetic code, discovered throughout the course of several ingenious experiments in the late 1950s, details the translation of an RNA molecule called messenger RNA (mRNA) into amino acids for protein creation. The apparent difficulty in translation is that somehow 4 RNA bases must be translated into a language of 20 amino acids; for every possible amino acid to be created, we must translate 3-nucleobase strings (called codons) into amino acids. Note that there are $4^3 = 64$ possible codons, so that multiple codons may encode the same amino acid. Two special types of codons are the start codon (AUG), which codes for the amino acid methionine, always indicates the start of translation, and the three stop codons (UAA, UAG, UGA), which do not code for an amino acid, cause translation to end.

The notion that protein is always created from RNA, which in turn is always created from DNA, forms the central dogma of molecular biology. It does not always hold; however, it offers an excellent approximation of the truth.

An organelle called a ribosome creates peptides by using a helper molecule called transfer RNA (tRNA). A single tRNA molecule possesses a string of three RNA nucleotides on one end (called an anticodon) and an amino acid at the other end. The ribosome takes an RNA molecule transcribed from DNA, called messenger RNA (mRNA), and examines it one codon at a time. At each step, the tRNA possessing the complementary anticodon bonds to the mRNA at this location, and the amino acid found on the opposite end of the tRNA is added to the growing peptide chain before the remaining part of the tRNA is ejected into the cell, and the ribosome looks for the next tRNA molecule.

Not every RNA base eventually becomes translated into a protein, and so an interval of RNA (or an interval of DNA translated into RNA) that does code for a protein is of great biological interest; such an interval of DNA or



RNA is called a gene. Because protein creation drives cellular processes, genes differentiate organisms and serve as a basis for heredity, or the process by which traits are inherited.

PROBLEM

The 20 commonly occurring amino acids are abbreviated by using 20 letters from the English alphabet (all letters except for B, J, O, U, X, and Z). Protein strings are constructed from these 20 symbols. Henceforth, the term genetic string will incorporate protein strings along with DNA strings and RNA strings.

The RNA codon table dictates the details regarding the encoding of specific codons into the amino acid alphabet.

Given: A file containing an RNA string s corresponding to a strand of mRNA (of length at most 10 kb).

Return: The protein string encoded by s .

Sample Dataset

```
AUGGCCAUGGCGCCCAGAACUGAGAUCAAUAGUACCCGUAUUAACGGGUGA
```

Sample Output

```
MAMAPRTEINSTRING
```

Hint: There are two main strategies to solve this problem. You can use the traditional strategy that the previous problems have introduced. This strategy involves making a dictionary representing the genetic code and using a **for** loop to convert the input RNA string into an output protein string.

However, you may also use an alternative strategy that leverages Biopython, a set of freely available tools for computational biology that are written in Python. Many of the homework problems can be solved more efficiently using Biopython functions and methods. Detailed instructions for installing and using BioPython are available [here](#).

Biopython offers a specific data structure called **Seq** for representing sequences. **Seq** represents an extension of the "str" (string) object type that is built into Python by supporting additional biologically relevant methods like **translate()** and **reverse_complement()**.

In this problem, you can easily use the **translate()** command. This strategy allows the problem to be solved in as little as four lines of code. Please let me know if you have problems getting Biopython up and running, as it will be referenced throughout the semester.